

HIGH-DIMENSIONAL PROBABILITY AND APPLICATIONS IN DATA SCIENCE

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LECTURE 1: THE CURSE OF DIMENSIONALITY AND THE MONTE CARLO METHOD

High-dimensional probability studies random objects with many coordinates. The key distinction is between many observations and many coordinates per observation. A method that works well with more data points may become too expensive when each point has many coordinates. Random sampling can help.

1.1 Big Data and High Dimension

A data set can be large in two independent ways. It may contain many data points, or each data point may itself have many coordinates. A list of incomes for a large population has many observations but only one coordinate per observation. In contrast, a 100×100 grayscale image is naturally a vector with 10^4 coordinates, one for each pixel. Text, sound, video, genomic data, medical histories, and game records give further examples of high-dimensional observations.

The number of observations often appears linearly in an algorithm or estimator. Dimension is more dangerous: a procedure that resolves every coordinate separately may require a number of operations exponential in the dimension.

Definition 1.1 [CURSE OF DIMENSIONALITY]. The curse of dimensionality means that the work or data needed for a problem can grow exponentially with the number of coordinates. Numerical integration on a grid, below, is the basic example.

1.2 Deterministic Integration on a Grid

Consider numerically approximating the integral of a function $f : [0, 1]^d \rightarrow \mathbb{R}$:

$$I(f) = \int_{[0,1]^d} f(x) dx.$$

In one dimension, a Riemann sum evaluates f on a grid. If the spacing between adjacent points is ε , then the interval needs approximately $1/\varepsilon$ points. Applying the same idea independently to d coordinates produces a Cartesian grid (figure 1) with

$$N \approx \left(\frac{1}{\varepsilon}\right)^d$$

points. Even a moderate resolution becomes infeasible when d is large. For example, ten points per coordinate require 10^d function evaluations.

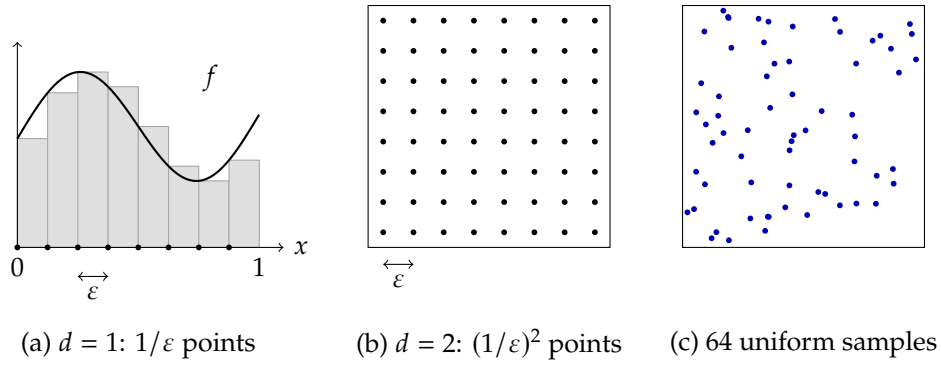


Figure 1: Grid-based and random evaluation points. (a) A Riemann sum evaluates f at points spaced ε apart. (b) The same resolution in every coordinate of $[0, 1]^2$ needs $(1/\varepsilon)^2$ points, and $(1/\varepsilon)^d$ in $[0, 1]^d$. (c) Monte Carlo instead evaluates f at independent uniform points, which make no attempt to cover the square.

The same phenomenon is visible geometrically (figure 2). The unit cube $[0, 1]^d$ has volume 1, while doubling every side produces a cube of volume

$$\text{vol}([0, 2]^d) = 2^d.$$

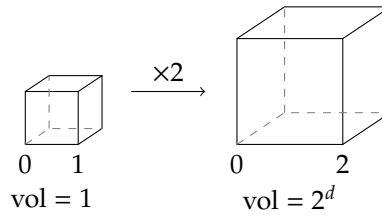


Figure 2: Doubling every side of the unit cube multiplies its volume by 2^d , drawn here for $d = 3$.

There is exponentially more volume to cover as dimension grows. A regular grid pays for this expansion by placing points in every combination of coordinate locations.

Note. A grid includes every combination of coordinate values. With ten choices in each of d coordinates, it needs 10^d points. That multiplication is why the grid grows so quickly.

1.3 Random Sampling

Probability offers a different approximation. Rather than cover the cube with a regular grid, draw independent random points

$$X_1, \dots, X_n \sim \text{Unif}([0, 1]^d)$$

and average the observed function values.

Definition 1.2 [MONTE CARLO ESTIMATOR]. The Monte Carlo estimator of $I(f)$ based on n independent uniform samples is

$$\widehat{I}_n(f) = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

Three probability facts explain why this works. If a random vector X has density p on $\mathbb{R}^d$, then

$$\mathbb{E}f(X) = \int_{\mathbb{R}^d} f(x)p(x) dx.$$

Expectation is linear, so $\mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}X_1 + \cdots + \mathbb{E}X_n$ without any independence assumption. Variances add when the random variables are independent:

$$\text{Var}(X_1 + \cdots + X_n) = \sum_{i=1}^n \text{Var}(X_i).$$

Finally, the strong law of large numbers states that if $Y_1, Y_2, \dots$ are independent and identically distributed with $\mathbb{E}|Y_1| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n Y_i \longrightarrow \mathbb{E}Y_1 \quad \text{almost surely.}$$

ALMOST-SURE CONVERGENCE means convergence with probability one: the outcomes on which the averages fail to converge have probability zero.

1.4 Unbiasedness and Convergence

The density of a uniform random vector X on $[0, 1]^d$ is 1 inside the cube and 0 outside. Consequently,

$$\mathbb{E}f(X) = \int_{[0,1]^d} f(x) dx = I(f).$$

Linearity of expectation then gives

$$\mathbb{E}\widehat{I}_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}f(X_i) = I(f).$$

Thus the Monte Carlo estimate is UNBIASED: its expected value is the true integral. If $f(X)$ is integrable, the strong law also gives

$$\widehat{I}_n(f) \longrightarrow I(f) \quad \text{almost surely.}$$

The variance tells us how much the estimate typically differs from the integral.

Proposition 1.3 [MEAN-SQUARE ERROR OF MONTE CARLO INTEGRATION]. *Suppose $X_1, \dots, X_n$ are independent and uniform on $[0, 1]^d$, and $\text{Var}(f(X_1)) < \infty$. Then*

$$\mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] = \frac{\text{Var}(f(X_1))}{n}.$$

Proof. Unbiasedness implies that the mean-square error equals the variance of the estimator. Independence and identical distribution yield

$$\begin{aligned} \mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] &= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n f(X_i) \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(f(X_i)) \\ &= \frac{\text{Var}(f(X_1))}{n}. \end{aligned}$$

□

If $|f(x)| \leq 1$ throughout the cube, then $\text{Var}(f(X_1)) \leq \mathbb{E}f(X_1)^2 \leq 1$. The root-mean-square error therefore satisfies

$$\left(\mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] \right)^{1/2} \leq \frac{1}{\sqrt{n}}.$$

The bound contains no explicit dependence on d . Achieving root-mean-square error at most ε requires $n \geq \varepsilon^{-2}$ samples, compared with roughly ε^{-d} grid points for coordinate-wise resolution ε . For $|f| \leq 1$, the sampling-error bound is independent of dimension. The grid comparison concerns spacing, not the same error guarantee; the cost of sampling a point and evaluating f may still depend on d .

Note. Monte Carlo points do not attempt to cover the cube (figure 1, right). They turn integration into estimation of an expectation, so independence controls error through variance rather than geometric coverage.

Source: Roman Vershynin, *High-Dimensional Probability and Applications in Data Science*, Lecture 1, “The Curse of Dimensionality and the Monte Carlo Method,” official course page

<https://www.math.uci.edu/~rvershyn/teaching/hdp/hdp.html> and lecture notes

<https://www.math.uci.edu/~rvershyn/teaching/hdp/Lecture1.pdf>.